LIGHTYEAR

MS90 Versioned judgement without rewriting history

Lightyear | Milestone report | 27 September 2026

Word edition

Outcome

The retained MS89 partial-invoicing observations pass the explicitly scoped version-2 comparison. The original MS89 run remains failed under its original judge. Both results are retained, linked by hashes, and answerable to a named rule set.

Customer claim: One agent-generated partial-invoicing journey has native Oracle/PostgreSQL evidence that satisfies the version-2 contract: exact numeric values for three named fields and accepted second-level shipment dates. This is a reassessment of existing evidence, not a new native run or full application qualification.

Evidence	Preserved or new result
Original MS89 campaign	`halted-nonrepairable`; 12 client calls and five native attempts retained
Original final native receipt	`failed`; original gate remains `passed: false`
Original comparison	`partial-invoicing-review-required`
New judge version	`idempiere-partial-invoicing-comparison-v2`
New assessment	`passed-bounded-partial-invoicing-contract-v2`
Additional execution	Zero model calls; zero native executions; no cloud resources started

Why this matters

A migration can produce the right business result while two engines print a number differently or store different time precision. A blanket normalization can hide a real defect. Leaving every representational difference unresolved can also obscure a valid result. MS90 makes the accepted boundary explicit, narrow, versioned and auditable.

The factory generated the candidate. The original deterministic judge exposed the differences. A person accepted a limited precision contract. A separately versioned deterministic judge then assessed the retained observations against that contract. The builder did not receive expected business outputs or edit its candidate during this reassessment.

The observed business journey

Both native lanes completed the same partial-invoicing scenario. Three ordered units were shipped and invoiced in portions of one and two. The observed invoice totals were 19.35 and 38.69, including tax of 1.35 and 2.70. The original lane checks verified 28 accounting entries in 12 balanced groups per engine. Starting stock of ten became seven. These business checks and expected values were not relaxed.

The precise comparison contract

Numeric values: equality without rounding

Only `order.discount`, `order.reserved` and `shipment.delivered` receive the new numeric rule. Both strings must parse as finite decimal values, and their mathematical values must be exactly equal. The comparison uses decimal arithmetic, not binary floating point. It introduces no tolerance and performs no rounding.

Field	Oracle	PostgreSQL	Version-2 treatment
`order.discount`	`10`	`10.00`	Equal exact decimal values
`order.reserved`	`3`	`3.0`	Equal exact decimal values
`shipment.delivered`	`3`	`3.0`	Equal exact decimal values

For example, `10` and `10.000000000000001` still differ. `NaN`, infinity, missing values and whitespace-padded values are rejected. The rule is not applied globally to other amounts, invoice totals or arbitrary trace fields.

Shipment dates: an owned precision decision

The signed decision `idempiere-oracle-date-seconds-v1` belongs to **Howard Weale**, takes effect on **26 September 2026**, and must be reviewed on **25 December 2026**. A new assessment on or after that review date is rejected until an appropriate renewed decision and versioned policy are supplied.

Its only column is `adempiere.m_inout.shipdate`, for the first and second shipments. The retained native catalogs and states show Oracle `DATE`, PostgreSQL `timestamp without time zone`, and UTC capture sessions on both sides. Each trace value must bind to the corresponding native shipment row and identifier.

The observed Oracle value is `2026-09-26T12:34:56`; PostgreSQL retains `2026-09-26T12:34:56.123456`. Version 2 admits that fractional loss only because the Oracle fraction is zero and every component through whole seconds matches. It never rounds into the next second.

Boundaries that remain enforced

A different second, another column, a different native datatype, an unsupported timezone context, an invalid date, an offset-bearing value or a mismatched row identity is not admitted. Subsecond ordering, audit timing and service-level timing guarantees remain outside the accepted scope. The captured database-session facts do not establish arbitrary application-client or production timezone equivalence.

Two native row differences and two corresponding trace differences describe the same two shipment timestamps. Along with the three numeric trace differences, the new report admits five trace findings and two row witnesses. These are not seven independent defects. Every original value remains visible.

Preservation and verification

One evidence set, two labelled verdicts

The assessment command verifies the source run's signatures, authorization, journal chain, plan, cleanup and implementation hashes. It hashes the retained source files, copies them to a separate

workspace, and reruns the unchanged original judge on that copy. This is a deterministic replay over recorded observations; no database is started.

The reproduced original gate must equal the original gate exactly. Only then can the new comparison run. Afterward, the command hashes the source files again and checks that the policy, decision and pinned implementation did not change. It refuses an existing output directory, keeping each publication separate.

The signed version-2 receipt reports both `original_gate_passed: false` and its own `passed: true`, together with `original_gate_reproduced: true` and `original_source_preserved: true`. The old run, original gate modules, candidate, observations and campaign accounting remain intact.

Evidence identities

Native run:

`journey-2ba2716e5c1749f89bdecd4cb1d17280`

Original native receipt SHA-256:

`48050fbc9b40cc011ef6af9a8b0beae434250a8132e9c02c15254ae11492638e`

Original gate SHA-256:

`0daa3611ff724b799782466ff74e1c5f9e7790162bc1934185edf6968689d7b2`

Version-2 comparison SHA-256:

`bcd6e62cd09ed800e02e5ad1bae2f57b480e2116a4ee1d6091a60e8ed7dcf6bf`

Version-2 signed receipt SHA-256:

`dde77f010e9f6dcd56071497706b1b201c84012a588cd4153a692ec2fe55f1fb`

What offline verification establishes

The published assessment contains the new plan and authorization, policy, signed timestamp decision, original receipt and journal, original gate and effect summary, native lane summaries, comparison and public key. The verification command checks the signatures and hash bindings and recomputes version 2 from these retained summaries.

That small publication does not contain all native row captures. Its verifier explicitly returns `full_native_gate_replayed: false`; the full original gate was replayed during assessment creation using the retained local observations. Local signatures provide attribution and tamper evidence, not independent attestation. The public key must be checked against the separately trusted operator identity.

Measured effort and validation

Preserve the complete cost of reaching this result

The table separates the cost already incurred by the MS89 campaign from the incremental deterministic assessment. Failed attempts remain included. The new scoped verdict does not retrospectively turn MS89's five original failed attempts into successful runs.

Measurement	Recorded value
MS89 client calls	12: five builder and seven analyst; 20 authorized
MS89 input / output tokens	248,122 / 24,044; zero cached input
MS89 native attempts	Five; final attempt completed both lanes but failed the old combined gate

Measurement	Recorded value
MS89 failed client invocations	Zero; usage available for all calls
MS89 agent / native time	804.031 / 1,488.561 seconds
MS89 execution-segment time	2,334.264 seconds (38.9 minutes), excluding stopped approval gaps
Human-authored candidate or repair-message bytes	Zero, provenance verified
Human controller assistance	Three disclosed MS89 controller revisions; new MS90 judge authored separately
MS90 recorded assessment time	52.969 seconds; excludes development, tests and document production
MS90 additional calls / native runs	Zero / zero
Dollar cost	Per-call billing unavailable; local energy unmetered; no GCP resources started

Zero repair bytes is a defined metric for candidate source and repair messages after the initial prompt. It does not mean that people wrote no specification, controller, judge or acceptance policy. MS90 is explicitly a human-authorized judge revision, not an autonomous semantic repair.

Validation completed

The original native gate was reproduced exactly from the retained observations, and every tracked source-evidence file remained unchanged. The signed publication verified successfully and recomputed the same version-2 comparison.

Eighteen focused tests cover the retained evidence result, exact decimal equality, changed numeric values, equal nonfinite values, unknown fields, timestamp seconds and offsets, column and datatype scope, UTC capture context, state hash binding, row-to-trace identity, Oracle fractional rejection, review-date expiry, business-check regression, policy expansion, signed publication replay and artifact tampering.

These tests check that the rule admits its intended cases and refuses nearby invalid cases. They are verification tests, not additional native customer journeys. Broader application, schema and platform qualification remain false.

Operating the assessment

Verify the published result

From the repository root, with the project environment installed, run:

```
$env:PYTHONPATH='src'
python tools/evaluate_journey_judge_v2.py verify `
  --output docs/calibration/idempiere-judge-v2
```

The expected status is `verified-versioned-assessment`, with `passed: true`, `original_gate_passed: false` and `full_native_gate_replayed: false`. Read the last field literally: this command verifies the published summaries and their signed lineage.

Reassess retained native observations

For a fresh, separately signed assessment, the operator must have the original local run, trusted signing material, unchanged pinned implementation and an effective timestamp decision. Use a new output path:

```
$env:PYTHONPATH='src'
$run='factory/idempiere/ms86-journeys/runs/'
$run+='journey-2ba2716e5c1749f89bdecd4cb1d17280'
python tools/evaluate_journey_judge_v2.py evaluate `
  --run $run `
  --output work/ms90/new-assessment
```

This copies the retained evidence, reruns the original deterministic gate, applies version 2 and signs a new assessment. An existing output directory is refused. Do not use the original verifier directly on the source run for this task: it writes derived summaries. Do not edit an old receipt to make its status pass.

Reporting and next work

Present this result as **bounded partial-invoicing contract equivalence, version 2, over retained native evidence**. Preserve the original failed run alongside it. Control Tower's original run status remains the historical result; this milestone adds a separate signed assessment artifact rather than replacing that status.

A future factory execution should pin this comparison policy before generation and native execution, then publish both its native outcome and the rule version. Further journeys should cover credit notes, reversals, concurrency and failure recovery. The timestamp owner must review the accepted precision scope by 25 December 2026. None of these future actions is implied by the reassessment.

Repository references

- Signed version-2 assessment
- Version-2 comparison with raw differences
- Preserved original gate
- Signed timestamp decision
- MS89 campaign receipt and costs
- MS89 milestone

Builder input provenance

There was no new builder input, generation or native run. This milestone reused MS89's final candidate and captured observations. The numeric and timestamp comparison policy came from MS89's exposed representation differences and the separately approved precision decision. A retrospective passing assessment is not a first-try generation result.